

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA 06
LEARNING

COURSE NAME: MACHINE

COURSE OUTCOME COVERED

CO1: Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces&Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation&Heuristic Search (BL1, BL2)

Assessment Tool Weightage: 20%

QUESTIONS: 15
Marks: 25

Total

Annexure A

CONCEPT MAPPING

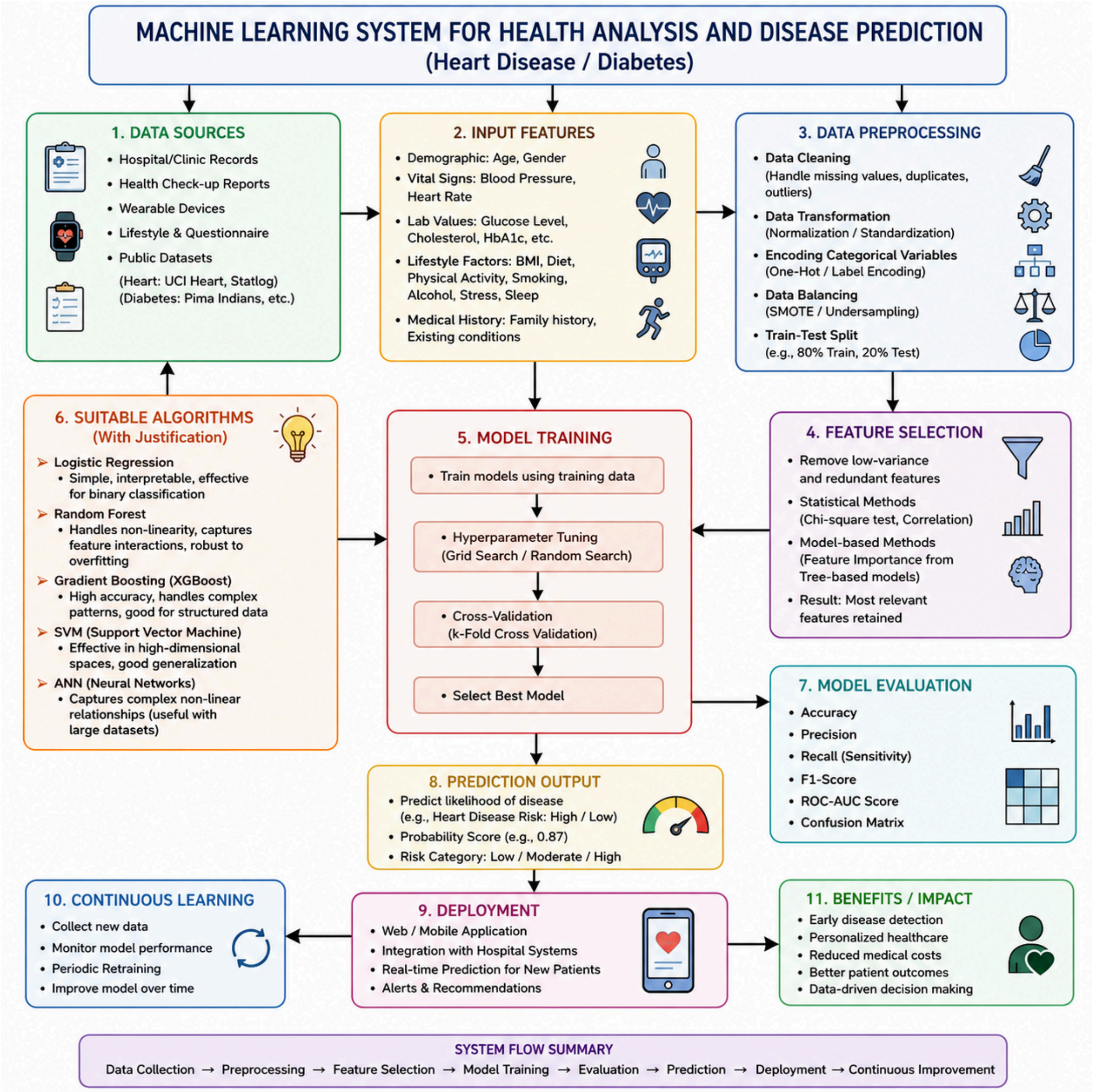
Code of Conduct:

I Anirudhan S / 192312056 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Anirudhan S

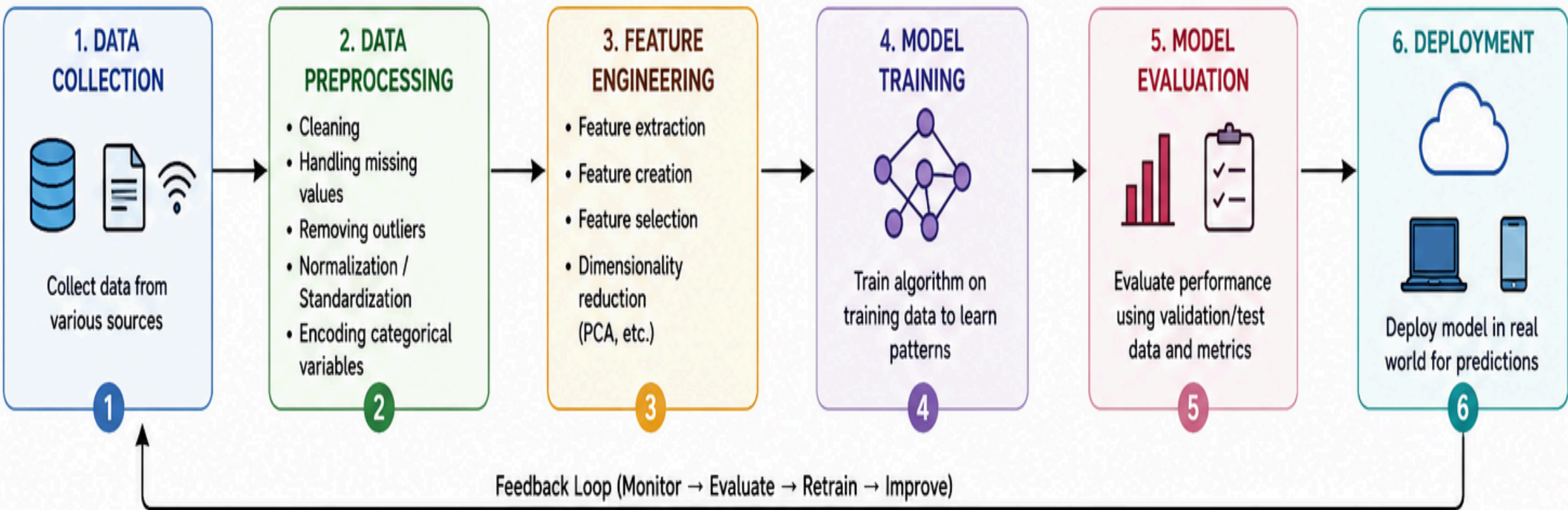
10. Design a machine learning system for health analysis to predict diseases such as heart disease or diabetes using patient data including age, blood pressure, glucose level, and lifestyle factors. Explain the choice of suitable algorithms and justify their selection. Describe the steps involved in data preprocessing, feature selection, and model training to ensure accurate and reliable predictions. I want concept map



11. Explain the architecture of a typical machine learning pipeline with a neat diagram. Discuss the role of each stage including data collection, preprocessing, feature engineering, model training, evaluation, and deployment. Also, differentiate between training error and generalization error with suitable examples.

MACHINE LEARNING PIPELINE: ARCHITECTURE, STAGES & ERROR ANALYSIS

1. ARCHITECTURE OF A TYPICAL ML PIPELINE

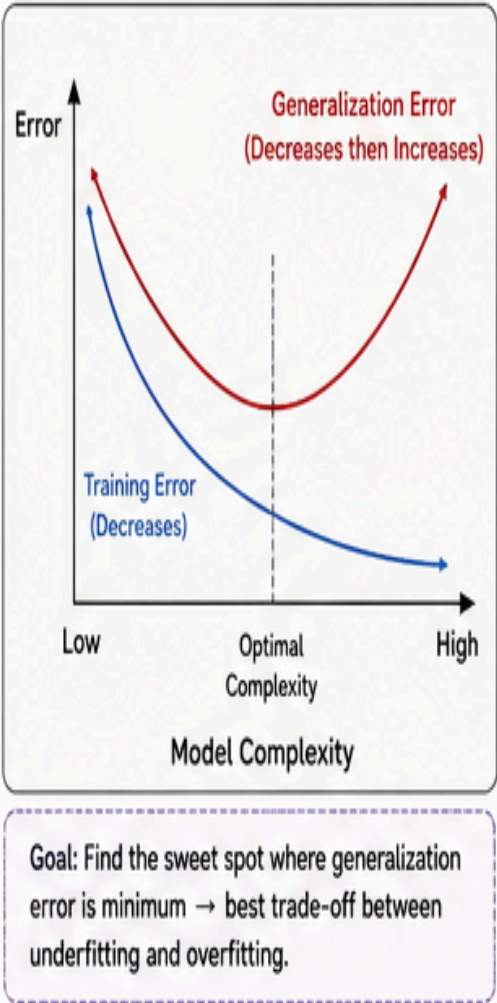


2. ROLE OF EACH STAGE

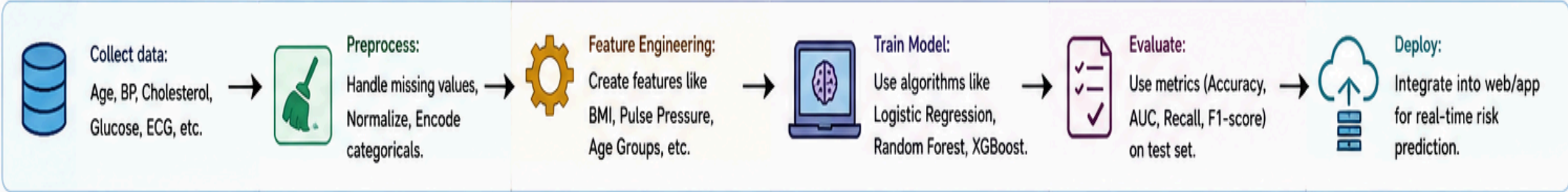
1. Data Collection	Gather relevant, diverse and high-quality data from databases, sensors, APIs, surveys, logs, etc. The quality of data directly impacts model performance.
2. Data Preprocessing	Prepare raw data to make it suitable for modeling. Removes noise, handles inconsistencies, converts data to a proper format and scales features to improve model accuracy and stability.
3. Feature Engineering	Create new features or transform existing ones that better represent underlying patterns. Helps in improving model performance and reducing complexity.
4. Model Training	Use training data to learn the relationship between input features and target variable by optimizing model parameters using algorithms.
5. Model Evaluation	Assess how well the model performs on unseen data using metrics like Accuracy, Precision, Recall, F1-score, RMSE, AUC, etc. Helps in model selection and tuning.
6. Deployment	Integrate the model into applications or systems to make predictions on new data. Monitor performance in production and update model as needed.

3. TRAINING ERROR vs GENERALIZATION ERROR

Training Error (In-sample Error)	Generalization Error (Out-of-sample Error)
- Error measured on the training data. - Indicates how well the model fits the training data it has seen. - Always decreases (or stays same) as model complexity increases. - Too low training error can indicate overfitting.	- Error measured on new, unseen data (validation/test set). - Indicates how well the model performs on real-world data. - Initially decreases with model complexity but after a point starts increasing (overfitting). - Lower generalization error means better real-world performance.
Formula: $E_{train} = \frac{1}{n} \sum_{i=1}^n L(y_i, \hat{y}_i)$ Example: A decision tree with very deep branches may predict training data almost perfectly ($E_{train} \approx 0$) but perform poorly on new data.	Formula: $E_{gen} = \frac{1}{m} \sum_{j=1}^m L(y_j, \hat{y}_j)$ Example: A model that memorizes training data will have high E_{gen} because it fails to generalize to unseen patients/customers.



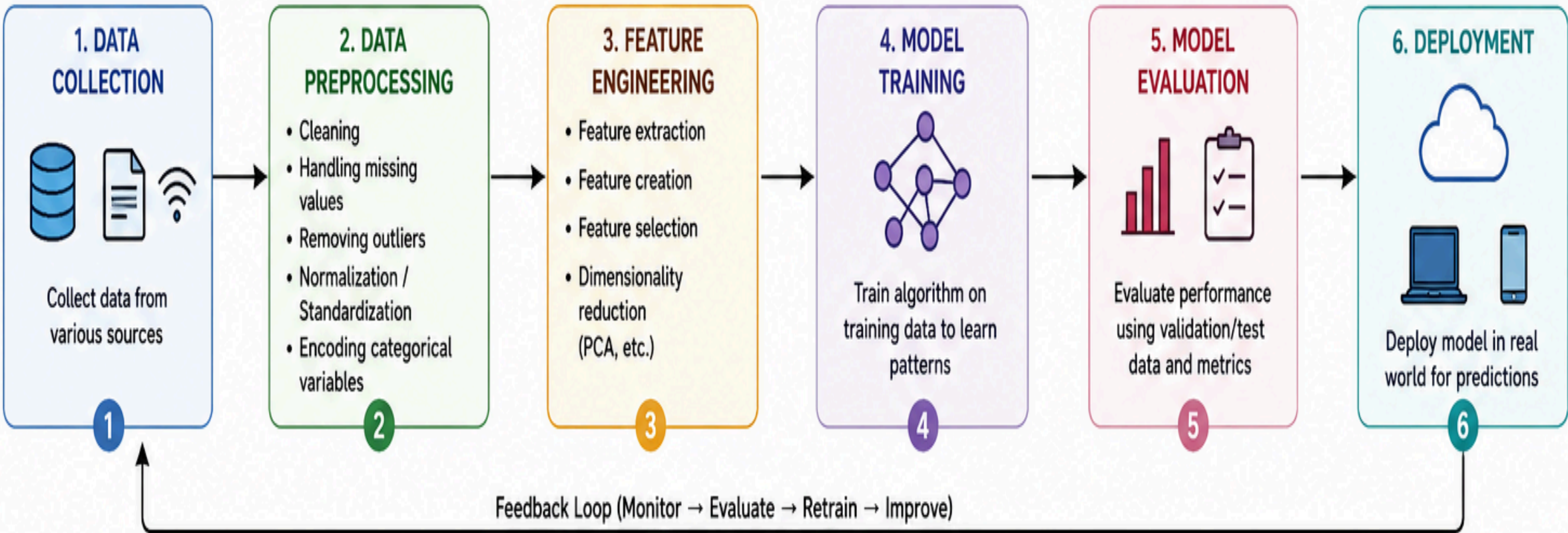
4. EXAMPLE OF PIPELINE (HEART DISEASE PREDICTION)



12. Design a machine learning model for spam email classification using labeled data. (5 Marks)(a) Explain the choice of algorithm (e.g., Naïve Bayes, SVM, or Logistic Regression).(b) Describe the preprocessing steps such as tokenization, stop-word removal, and vectorization.(c) Discuss evaluation metrics such as accuracy, precision, recall, and F1-score.

MACHINE LEARNING PIPELINE: ARCHITECTURE, STAGES & ERROR ANALYSIS

1. ARCHITECTURE OF A TYPICAL ML PIPELINE



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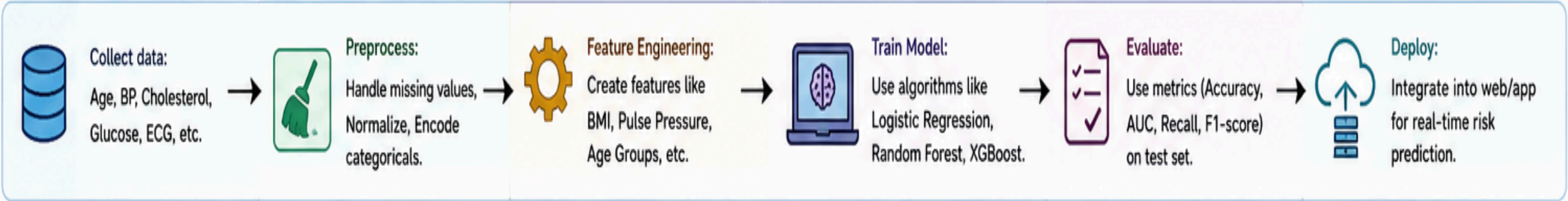
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Goal: Find the sweet spot where generalization error is minimum → best trade-off between underfitting and overfitting.

4. EXAMPLE OF PIPELINE (HEART DISEASE PREDICTION)



13. In the context of the Candidate-Elimination Algorithm, consider a dataset with attributes: (5 Marks)Sky, AirTemp, Humidity, Wind, Water, Forecast and target concept EnjoySport.(a) Explain the initialization of S (specific boundary) and G (general boundary).(b) Show how S and G change with each training example.(c) Compare Candidate-Elimination with the Find-S algorithm in terms of generalization capability.

13. Candidate-Elimination Algorithm

Dataset (Attributes: Sky, AirTemp, Humidity, Wind, Water, Forecast → Target Concept: EnjoySport)

Example	Sky	AirTemp	Humidity	Wind	Water	Forecast	EnjoySport
x_1	Sunny	Warm	Normal	Strong	Warm	Same	Yes
x_2	Sunny	Warm	High	Strong	Warm	Same	Yes
x_3	Rainy	Cold	High	Strong	Warm	Change	No
x_4	Sunny	Warm	High	Strong	Cool	Change	Yes
x_5	Sunny	Cold	Normal	Strong	Cool	Change	No
x_6	Rainy	Warm	Normal	Strong	Warm	Same	Yes

(a) INITIALIZATION OF S AND G

1. Specific boundary (S):

Most specific hypothesis (contains only one hypothesis)
All attributes are set to the most specific value Φ (null symbol).

$S_0 = \langle \Phi, \Phi, \Phi, \Phi, \Phi, \Phi \rangle$

2. General boundary (G):

Most general hypothesis (set of hypotheses)
Each attribute can take any value from its domain.

$G_0 = \{ \langle ?, ?, ?, ?, ?, ? \rangle \}$

where ? represents "any value".

Legend

Φ : Most specific value (no value)
? : Any value

This ensures that

$S_0 \subseteq C \subseteq G_0$

where C is the set of all candidate hypotheses consistent with training examples seen so far.

(b) CHANGE OF S AND G WITH EACH TRAINING EXAMPLE

Step	Training Example (x_i)	Class (EnjoySport)	Specific Boundary (S)	General Boundary (G)
0 (Init.)	–	–	$\langle \Phi, \Phi, \Phi, \Phi, \Phi, \Phi \rangle$	$\{ \langle ?, ?, ?, ?, ?, ? \rangle \}$
1	$x_1 = \langle \text{Sunny, Warm, Normal, Strong, Warm, Same} \rangle$	Yes	$\langle \text{Sunny, Warm, Normal, Strong, Warm, Same} \rangle$	$\{ \langle ?, ?, ?, ?, ?, ? \rangle \}$
2	$x_2 = \langle \text{Sunny, Warm, High, Strong, Warm, Same} \rangle$	Yes	$\langle \text{Sunny, Warm, ?, Strong, Warm, Same} \rangle$	$\{ \langle ?, ?, ?, ?, ?, ? \rangle \}$
3	$x_3 = \langle \text{Rainy, Cold, High, Strong, Warm, Change} \rangle$	No	$\langle \text{Sunny, Warm, ?, Strong, Warm, Same} \rangle$	$\{ \langle \text{Sunny, ?, ?, ?, ?, ?} \rangle, \langle ?, \text{Warm, ?, ?, ?, ?} \rangle, \langle ?, ?, ?, ?, \text{Same} \rangle \}$
4	$x_4 = \langle \text{Sunny, Warm, High, Strong, Cool, Change} \rangle$	Yes	$\langle \text{Sunny, Warm, ?, Strong, ?, ?} \rangle$	$\{ \langle \text{Sunny, ?, ?, ?, ?, ?} \rangle, \langle ?, \text{Warm, ?, ?, ?, ?} \rangle \}$
5	$x_5 = \langle \text{Sunny, Cold, Normal, Strong, Cool, Change} \rangle$	No	$\langle \text{Sunny, Warm, ?, Strong, ?, ?} \rangle$	$\{ \langle \text{Sunny, Warm, ?, ?, ?, ?} \rangle \}$
6	$x_6 = \langle \text{Rainy, Warm, Normal, Strong, Warm, Same} \rangle$	Yes	$\langle ?, \text{Warm, ?, Strong, ?, ?} \rangle$	$\{ \langle \text{Sunny, Warm, ?, ?, ?, ?} \rangle \}$

Final Version Space:

$S_{final} = \langle ?, \text{Warm, ?, Strong, ?, ?} \rangle$

$G_{final} = \{ \langle \text{Sunny, Warm, ?, ?, ?, ?} \rangle \}$

All hypotheses between S_{final} and G_{final} (inclusive) are consistent with all training examples.

(c) COMPARISON WITH FIND-S ALGORITHM

Aspect	Candidate-Elimination	Find-S
Approach	Maintains both a specific boundary (S) and a general boundary (G) to represent the version space.	Maintains only the most specific hypothesis that is consistent with all positive examples.
Hypotheses Maintained	Set of all hypotheses between S and G ($S \subseteq C \subseteq G$).	Single hypothesis h (most specific).
Use of Negative Examples	Uses both positive and negative examples to generalize S and specialize G.	Uses only positive examples; ignores negative examples.
Generalization Capability	More general. Considers boundaries of the concept, so can converge to a more accurate version space.	Less general. May produce overly specific hypothesis (overfitting).
Convergence Speed	Usually slower (maintains sets and updates both boundaries).	Faster (updates a single hypothesis).
Output	Version space (set of possible hypotheses).	Single hypothesis.
Risk of Overfitting	Lower (because negative examples restrict G).	Higher (cannot exclude unseen negative cases).



Key Takeaway:

Candidate-Elimination is more powerful than Find-S because it uses both positive and negative examples and returns the entire hypothesis space, leading to better generalization.

Notation:

Φ – most specific value (no value)
? – any value

S – specific boundary

G – general boundary

C – candidate hypothesis space

14. Develop a machine learning system for predicting house prices using features like location, size, number of rooms, and amenities. (5 Marks)(a) Justify the choice of regression model (Linear Regression, Decision Tree, or RandomForest).(b) Explain feature scaling and handling missing data.(c) Describe how overfitting can be detected and avoided.

14. DEVELOP A MACHINE LEARNING SYSTEM FOR PREDICTING HOUSE PRICES

Goal: Predict the price of a house using features such as location, size, number of rooms, and amenities.

SYSTEM PIPELINE

1. Data Collection



- Historical house data
- Sources: Real estate portals, CSV datasets, APIs

2. Data Preprocessing



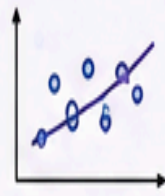
- Handle missing data
- Encode categorical variables
- Feature scaling
- Train-test split

3. Feature Engineering



- Create useful features (e.g., price per sq ft)
- Combine / transform features
- Remove irrelevant features

4. Model Training



- Choose regression model
- Train on training data
- Hyperparameter tuning (if needed)

5. Evaluation



- Evaluate on test data using metrics
- Compare models

6. Deployment



- Deploy the best model
- Web / App for price prediction
- Monitor performance

EXAMPLE FEATURES

Feature	Type	Description	Example
Location	Categorical	Area or neighborhood	Whitefield, Koramangala
Size (Sqft)	Numerical	Total built-up area	1500
No. of Rooms	Numerical	Total bedrooms	3
Bathrooms	Numerical	Number of bathrooms	2
Amenities	Categorical/Binary	Lift, Parking, Gym, Swimming Pool, etc.	1/0 (Yes/No)
Age of House	Numerical	Age in years	5
Price (Target)	Numerical	Selling price (in ₹)	85,00,000

Chosen Model: Random Forest Regressor

Justification:

- Captures non-linear relationships between features and price.
- Handles complex interactions between variables (e.g., location and amenities together).
- Robust to outliers and noise in data.
- Works well with both numerical and categorical features.
- Provides feature importance, which helps in interpretability.
- Generally gives higher accuracy than simple models like Linear Regression.
- Less prone to overfitting compared to a single Decision Tree.

Note: For small and linear datasets, Linear Regression is simple and interpretable. Decision Trees are easy to visualize but may overfit; Random Forest (ensemble of trees) gives better generalization and accuracy.

(a) JUSTIFY THE CHOICE OF REGRESSION MODEL

1. Handling Missing Data

Feature Type	Method
Numerical (e.g., Size, Age)	Impute with Mean / Median or using KNN Imputer
Categorical (e.g., Location)	Impute with Mode or create a new category "Unknown"
High % Missing	Drop the feature if missing values > threshold (e.g., 40%)

2. Feature Scaling

Scaling ensures that all features contribute equally to the model. Important for models like Linear Regression, SVR, k-NN.

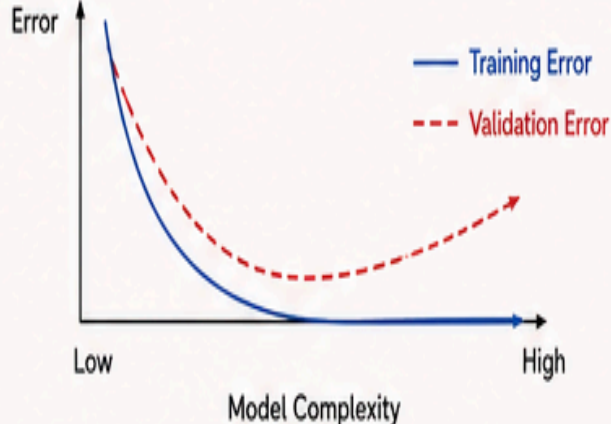
Method	Formula / Description
Standardization (Z-score)	$z = \frac{x - \mu}{\sigma}$ where μ = mean, σ = standard deviation
Min-Max Scaling (Normalization)	$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$ scales values to [0, 1] range

Note: Tree-based models (e.g., Random Forest, Decision Tree) do not require scaling but it still helps for consistency.

(c) DETECTION AND AVOIDANCE OF OVERFITTING

1. How to Detect Overfitting

- High performance on training data but low on test/validation data.
- Large gap between training error and validation error.
- Learning curve: training error decreases while validation error starts increasing.



2. How to Avoid Overfitting

- Cross-validation to evaluate model on multiple splits.
- Use simpler models / limit tree depth (for trees).
- Regularization (L1/L2) for Linear Regression / Ridge / Lasso.
- Hyperparameter tuning (e.g., n_estimators, max_depth in Random Forest).
- Increase training data if possible.
- Feature selection / remove noisy or irrelevant features.

EVALUATION METRICS

$MAE \text{ (Mean Absolute Error)} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$

$MSE \text{ (Mean Squared Error)} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$

$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$

$R^2 \text{ Score} = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$

SUMMARY

We use a regression model (preferably Random Forest) to predict house prices. Data is preprocessed (missing values handled, features scaled), model is trained and evaluated. Overfitting is detected using validation error and avoided using regularization, cross-validation, and tuning.

Rubrics For Calculation

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand